Midterm 2 Revision

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1. Consider the IBVP:

$$u_t - u_{xx} = f(t)\sin(3x),\ u(0,t) = u(\pi,t) = 0,\ u(x,0) = x - 1, \tag{1}$$

where $0 < x < \pi$, $t > 0$, $f(t) = 1$ for $0 \le t \le 1$, and $f(t) = 0$ for $t > 1$.

Proof (Hernandez). By superposition, $u = v + w$ where v and w are solutions to the following problems:

$$v_t - v_{xx} = f(t)\sin(3x),\ v(0,t) = v(\pi,t) = 0,\ v(x,0) = 0, \tag{2}$$
$$w_t - w_{xx} = 0,\ w(0,t) = w(\pi,t) = 0,\ w(x,0) = x - 1, \tag{3}$$

of the following form:

$$v(x,t) = \sum_{n=1}^{\infty} a_n(t)\sin(nx), \qquad w(x,t) = \sum_{n=1}^{\infty} b_n(t)\sin(nx),$$

because we obtain an SLP with the following eigenpairs:

$$\varphi_n(x) = -\lambda_n^2\varphi_n(x), \qquad \varphi_n(0) = \varphi_n(\pi) = 0,$$
$$\varphi_n(x) = \sin(\lambda_n x), \qquad \lambda_n = n.$$

Substituting into Equation (2) yields:

$$\sum_{n=1}^{\infty} \dot{a_n}(t)\sin(nx) - \sum_{n=1}^{\infty} -n^2 a_n(t)\sin(nx) = \sum_{n=1} \delta_{3,n} f(t)\sin(nx)$$
$$\dot{a_3}(t) + 3^2 a_3(t) = f(t), \qquad \dot{a_n}(t) + n^2 a_n(t) = 0, \qquad a_n(0) = 0, \qquad \forall n \neq 3,$$

and substituting into Equation (3) yields:

$$\sum_{n=1}^{\infty} \dot{b_n}(t)\sin(nx) - \sum_{n=1}^{\infty} -n^2 b_n(t)\sin(nx) = \sum_{n=1} 0\sin(nx)$$
$$\dot{b_n}(t) + n^2 b_n(t) = 0, \qquad b_n(0) = \beta_n = \frac{2}{\pi}\int_0^{\pi} (x-1)\sin(nx)\ dx = \frac{2(1-\pi)(-1)^n - 2}{n\pi},$$

where β_n are the Fourier coefficients of $w(x,0)$ and thus $b_n(t) = \beta_n e^{-n^2 t}$.
Lastly, $a_n(t) = 0$ for $n \neq 3$, and a_3 depends on t as follows:

$$a_3(t) = \int_0^t e^{-(t-s)}\ ds = 1 - e^{-t} \qquad \text{if } t \le 1,$$
$$a_3(t) = \int_0^1 e^{-(t-s)}\ ds = (e-1)e^{-t} \qquad \text{if } t > 1.$$

□

2. Consider the IBVP:

$$u_{tt} = u_{xx}, \quad 0 < t, x < 1;\ u_x(0,t) = 1,\ u_x(1,t) = 0,\ u(x,0) = 1,\ u_t(x,0) = 0. \tag{4}$$

Proof (Hernandez). The solution is given by a superposition $u = v + w$ whose terms solve the following problems:

$$v_{tt} = v_{xx};\ v_x(0,t) = 1,\ v_x(1,t) = 0,\ v(x,0) = 0,\ v_t(x,0) = 0. \tag{5}$$
$$w_{tt} = w_{xx};\ w_x(0,t) = 0,\ w_x(1,t) = 0,\ w(x,0) = 1,\ w_t(x,0) = 0, \tag{6}$$

whose first solution is given by $v(x,t) = x - t$ on $0 \le x \le t < 1$ and $v(x,t) = 0$ on $t < x < 1$. Both pieces are linear and thus satisfy the PDE. The first constraint is active on $0 \le x \le t < 1$, and satisfied because $v_x(0,t) = 1 \cdot x^{1-1} = 1$. The remaining constraints are active on $t < x < 1$, and are satisfied because v and v_x are identically zero.

The problem for w yields an SLP with the following eigenpairs:

$$\varphi_n(x) = -\lambda^2 \varphi_n(x), \qquad \varphi_(0) = \varphi(1) = 0,$$
$$\varphi_n(x) = \cos(\lambda_n x), \qquad \lambda_n = n.$$

I think those are supposed to be sin, but the solution says cos, so I'm having trouble figuring out what went wrong. □

3. When $0 < r < 1$ and $0 < t$, consider the IBVP below:

$$u_{tt} = \Delta u, \qquad u(1,t) = u|_{r=1} = 0,\ u(r,0) = u|_{t=0} = 0,\ u_t(r,0) = u_t|_{t=0} = r.$$

Proof (Hernandez). We investigate the situation when we're working (1) on a unit disk, and (2) on a unit ball.

(a) In the 2-d situation, our equation takes the following form:

$$u_{tt} = u_{rr} + \frac{1}{r} u_r.$$

As in Section 4.5, our solution has the following form:

$$u(r,t) = \sum_{n=1}^{\infty} c_n \frac{\sin(z_n t)}{z_n} J_0(z_n r),$$

where the coefficients c_n are determined from the following equations:

$$r = \sum_{n=1}^{\infty} c_n J_0(z_0 r),$$

and are thus given by:

$$c_n = \frac{\int_0^1 f(r) J_0(\frac{z_0}{1}) r \ dr}{||J_0(\frac{z_0}{1})||^2}.$$

(b) The general solution for the 3-d problem is given by:

$$u(r,t) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2T_0}{n} e^{-n^2 kt} \frac{\sin(nr)}{r}$$

□

4. Consider the wave equation in a ring:

$$u_{tt} = c^2 u_{xx},\ x \in (0,l), t > 0; u(0,t) = u(l,t)\ u_x(0,t) = u_x(l,t), \tag{7}$$

$$u(x,0) = u_0,\ u_t(x,0) = 0. \tag{8}$$

Proof (Hernandez). Consider the following superposition of odd periodic extensions of the constant initial condition:

$$u(x,t) = \frac{1}{2}\left(F(x-ct) + F(x+ct)\right) = \frac{1}{2}\left(u_0 + u_0\right) = \frac{2u_0}{2} = u_0, \qquad \text{where } F(\xi) = u_0.$$

In addition to $u(x,0) = u_0$, the solution also satisfies $u_{tt} = c^2 u_{xx} = 0$, $u(0,t) = u(l,t) = u_0$, $u_x(0,t) = u_x(l,t) = 0$, $u_x(x,0) = 0$ because $u(x,t)$ is a constant whose derivatives are thus zero.

The first row of boundary conditions implies the density is held fixed at its endpoints, and the second row of initial conditions implies the string starts with constant and unchanging density along the string. Therefore, the solution corresponds to a string with constant density u across space and time. □